

Chapter Two

Nuclear Weapons Primer

Nuclear Fission and Fusion

A nuclear weapon is a device in which most or all of the explosive energy is derived from either fission, fusion, or a combination of the two nuclear processes.²

(1) **NUCLEAR FISSION** is the splitting of the nucleus of an atom into two (or more) parts. Certain isotopes of uranium and plutonium (and some other heavier elements), when bombarded by neutrons, will split into atoms of lighter elements and in the process will emit, on average, two or more neutrons from each nucleus and considerable energy—about ten million times as much, atom for atom, as is obtained from ordinary chemical combustion.³

(2) **NUCLEAR FUSION** is the joining (or fusing) of the nuclei of two atoms to form a single heavier atom. At extremely high temperatures—in the range of tens of millions of degrees—the nuclei of isotopes of hydrogen (and some other light elements) can readily combine to form heavier elements and in the process release considerable energy.⁴ While a number of thermonuclear reactions are identified in the literature,⁵ the most relevant to nuclear weapons is the reaction between deuterium (H-2 or D) and tritium (H-3 or T), two hydrogen isotopes, because the deuterium-tritium (D-T) reaction proceeds more rapidly at realizable temperatures than the other fusion reactions.⁶

Atom for atom, the energy released in fusion is less than that released in fission,⁷ but the atoms involved in fusion are much lighter, so in theory, the maximum energy obtainable from fusion is about three or four times as great per unit weight as the maximum energy obtainable from nuclear fission.⁸ Pure fission weapons of low yield are the simplest practical nuclear weapons to design and construct. Some low yield weapons, e.g., older nuclear artillery shells, are the pure fission type. Weapons of very high yield and weapons that have the highest yield-to-weight ratio use a combination of fission and fusion reactions.⁹ Most weapons in the U.S. stockpile are probably of this type.

Fission Weapons

Chain Reaction

In a fission device (a weapon or a reactor), it is necessary to achieve a *chain reaction*, whereby neutrons emitted by fissioning nuclei induce fission in other fissionable nuclei. The neutrons from the fissions, in turn, induce fission in still other fissionable nuclei, and so on. When uranium-235 (U-235) fissions, an average of about 2.56 neutrons are released; an average of about 2.9 to 3.0 neutrons are released when a nucleus of plutonium-239 (Pu-239) fissions.¹⁰ A portion of these neutrons is captured by nuclei that do not fission, and others

1 This section is derived from Ted Greenwood, George W. Rathjens and Jack Ruina, "Nuclear Power and Weapons Proliferation" (London: International Institute for Strategic Studies, Adelphi Paper No. 130, Winter, 1976), pp. 2-6. Other useful primers, although not an exhaustive list, include: Samuel Glasstone and Philip J. Dolan, *The Effects of Nuclear Weapons*, 3rd ed. (Washington: U.S. DOD and U.S. DOE, 1977), Chapter 1; Philip Morrison, "The Physics of the Bomb," *Atomic Energy* (Harmondsworth, Middlesex, U.K.: Penguin Books, 1950), pp. 2-29, 101-125, 194-201; Edward Teller, et al., *The Constructive Use of Nuclear Explosives* (N.Y.: McGraw Hill, 1968), pp. 1-91; R. Serber, et al., "The Los Alamos Primer," LA-1, April 1943; Robert W. Selden, "An Introduction to Fission Explosives," LLL, UCID 15554, July 1969; J. Carson Mark, et al., "Nuclear Weapons," *Nuclear Proliferation and Safeguards* (N.Y.: Praeger Publishers, 1977), Chapter VI, pp. 139-145; M.B. Neyman and K.M. Sadilenko, "Thermonuclear Weapons," translation by Technical Information Center (Wright Patterson AFB, OH, October 1960); Torsten Magnusson, "Design and Effects of Atomic Weapons" (U.S. Joint Publication Research Service, 8295, 22 May 1961); Howard Morland, "The H-bomb Secret," *The Progressive*, November 1979, pp. 14-23; (read with "Errata," *The Progressive*, December 1979, p. 36) and *The Secret that Exploded* (New York: Random House, 1981); numerous encyclopedia articles by authors including Edward Teller, Hans Bethe, John S. Foster.

2 While all nuclear weapons deployed contain chemical high explosive material to initiate the nuclear reaction, it is theoretically possible to use electromagnetic energy (i.e., light from lasers) instead of chemical explosives.

3 Approximately 200 MeV (3.2×10^{-11} joules) is released in each fission, but only about 180 MeV is immediately available as explosive energy (from gamma rays and the kinetic energy of fission products and neutrons, and from only a small fraction of the decay energy of fission products). Based on the latter value, an explosion equivalent to 1 kiloton (Kt) of TNT (defined as the release of 10^{11} calories) is obtained by the fission of 1.45×10^{24} nuclei. Thus, the complete fission of one kg of fissionable material produces a 17.5 Kt explosion, or 1 Kt is released from the complete fission of 0.057 kg (57 grams or 2 ounces); see, Glasstone and Dolan, *op. cit.*, pp. 12-13.

4 The principal thermonuclear reactions involving isotopes of hydrogen include:

(i) $D + T \rightarrow He^4 (3.52 \text{ MeV}) + n (14.07 \text{ MeV})$
 (ii) $D + D \rightarrow \begin{cases} T (1.01 \text{ MeV}) + p (3.02 \text{ MeV}) \\ He^3 (0.82 \text{ MeV}) + n (2.45 \text{ MeV}) \end{cases}$
 (iii) $T + T \rightarrow He^4 + 2n + 11.4 \text{ MeV}$

The (D + D) reaction proceeds with approximately equal probability via the two channels.

5 See for example, Neyman and Sadilenko, *op. cit.*, p. 8.

6 The (D-T) reaction is reaction (i) in footnote 4. It is about 100 times more probable than the (D + D) reaction (ii) in the temperature range 10-100 KeV (1 KeV = 11.6 million "K"). Alternately, a given reaction rate can be achieved at a lower temperature for the (D-T) reaction than for other fusion fuels; Booth, et al., "Prospects of Generating Power with Laser-Driven Fusion," *Proceedings IEEE*, 64, October 1976, p. 1461.

7 17.6 MeV for the D-T reaction compared to 200 MeV for fission of a uranium or plutonium atom.

8 Compared to 200 MeV for fission, approximately four times for fusion of deuterium with tritium nuclei and approximately three times for fusion of deuterium nuclei or deuterium and tritium nuclei from lithium-6 deuteride; see Glasstone and Dolan, *op. cit.*, p. 21, for derivation.

9 Although possible in principal, pure fusion weapons—that is, where the high temperature necessary for fusion is not obtained from a fission explosion—have not as yet been developed in practical form and may never be. One of the objectives of the U.S. inertial confinement fusion research program probably is to determine the feasibility of using lasers to produce a practical pure fusion weapon.

10 These values are for fission induced by 1 MeV neutrons. The average number of neutrons per fission decreases slightly as the energy of the neutron inducing the fission drops; see USAEC, *Reactor Physics Constants*, ANL-6800, July 1963, pp. 20-23, and A.M. Weinberg and E.P. Wigner, *The Physical Theory of Neutron Chain Reactors* (Chicago: University of Chicago Press, 1958), p. 129.

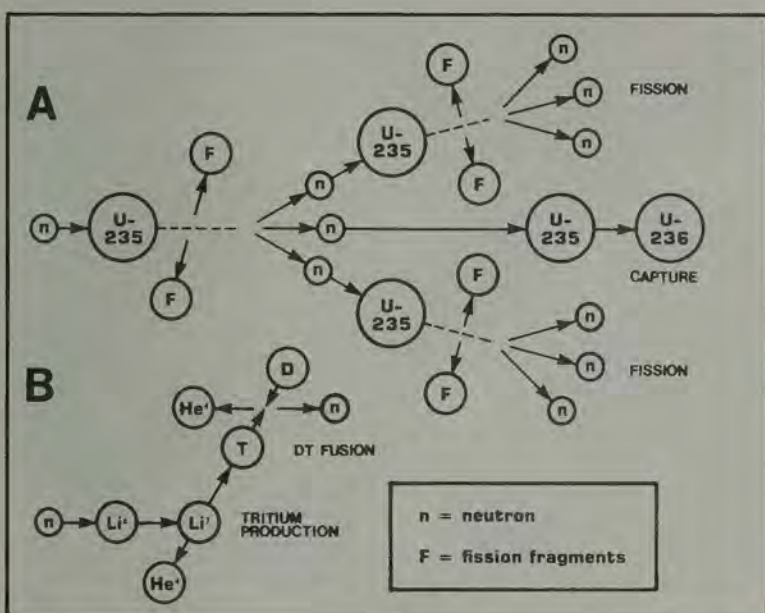


Figure 2.1 (a) Fission chain reaction in uranium-235. (b) Fusion of deuterium and tritium into helium [upper right]. Here, the tritium is produced by absorption of a neutron in lithium-6 (lower left).

escape the material without being captured. What is left can cause further fissions. If more than one neutron per fission remains for the chain reaction, more fissions are achieved in the next "generation" than in the previous one. To achieve a high efficiency in a nuclear explosion, a very rapid growth in the number of fissions is sought—that is, a rapidly multiplying chain reaction. This means, among other things, that an effort must be made to keep down the leakage of neutrons out of the fissile material and to avoid neutron absorbing impurities in the fissionable material.

Fissionable Materials

Many heavy atomic nuclei are capable of being fissioned; but only a fraction of these are fissile, which means fissionable by slow (or zero energy) neutrons, as well as fast (highly energetic) neutrons. Since the neutrons resulting from nuclear fission are emitted with a wide range of energies, nuclei which fission only from the capture of fast neutrons would generally not be able to sustain a chain reaction.¹¹ From a practical point of view, fission weapons must be made using fissile materials,¹² principally U-235, Pu-239, U-233, or some combination of these.

U-238 and Thorium-232 (Th-232), both abundant in nature, are also fissionable but only by fast neutrons, so they cannot sustain a chain reaction by themselves. Nevertheless, these two materials can contribute to the yield in both fission and thermonuclear explosions where the many excess high-energy neutrons generated by other fission and by fusion reactions can cause them to fission (see below: Fusion Weapon Design, Thermonuclear Weapon Design).

Uranium as found in nature consists primarily of two isotopes, U-235 and U-238, with U-235 (the fissile isotope) occurring only 0.711 percent in abundance, while U-238 constitutes 99.3 percent. Plutonium does not occur naturally except in minute concentrations. Therefore the fissile isotope Pu-239 is made artificially in nuclear reactors from U-238.¹³

To date only U-235 and Pu-239 seem to be used as the fissile material in stockpiled nuclear weapons. Other fissile isotopes of uranium and plutonium, e.g., U-233 and Pu-241, occur only in trace amounts in nuclear weapons. Because U-233 and Pu-241 are more difficult and costly to produce in quantity than U-235 and Pu-239, they are not used as primary fissile material. U-233 and Pu-241 are also more radioactive, can complicate weapons fabrication, and can degrade the reliability of other weapon components.¹⁴ U-238 is used to contribute to the yield of some fission and thermonuclear weapons; Th-232 does not appear to be used.

Fission weapons are made using grades of enriched uranium (from an enrichment plant) or plutonium (produced in a reactor) which contain the fissile isotopes U-235 and Pu-239, respectively, in levels which provide efficient reactions and use a minimum of materials. However, fission weapons do not require uranium or plutonium pure in the isotopes U-235 and Pu-239 to make an explosion, nor do they require uranium or plutonium in the form of a metal. Theoretically, uranium weapons cannot be made using uranium enriched to less than about 5-6 percent U-235. In the range 6-10 percent U-235, very large quantities—thousands of kilograms—of uranium would be required. The fissile uranium material of current nuclear weapons is 93.5 percent enriched U-235.¹⁵

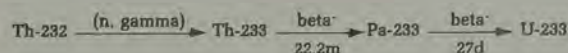
In theory plutonium compounds containing 6-10 percent (or even less) Pu-239 are usable for weapons. In practice, plutonium is produced from U-238 in a reactor

11 Generally these would include isotopes of heavy elements with an even isotope number, e.g., U-236; U-238. Pu-240 is an exception in that it is not fissile and yet can sustain a chain reaction with fast neutrons.

12 Generally these would include isotopes of heavy elements with an odd isotope number, e.g., U-235; Pu-239; Pu-241.

13 In the reaction: $\text{U-238} \xrightarrow{(n, \gamma)} \text{U-239} \xrightarrow{\beta^- 23.5m} \text{Np-239} \xrightarrow{\beta^- 2.35d} \text{Pu-239}$

14 U-233 is produced in nuclear reactors from Thorium-232 in the reaction:



The U-233 is used as the primary fuel in some types of nuclear reactors. Interest in U-233 as a nuclear weapons usable material stems from concern that, like highly enriched uranium (U-235) and plutonium, U-233 may be diverted from its use in civil activities and used in weapons. (The U.S. has tested U-233 weapons.)

15 SASC, FY 1983 DOD, Part 7, p. 4979.

to a purity of about 93.5 percent Pu-239 for weapons use.¹⁶ The element Pu-240, a byproduct of plutonium production, is an undesirable element for weapons design because of its high spontaneous fission rate.¹⁷ Consequently, in reactors used for the production of plutonium for weapons, the period of time that the U-238 is left in the reactor is restricted to limit buildup of Pu-240 (to about 6 percent) while creating the fissile Pu-239.

Although one basic difference between U-235 and Pu-239 for weapons design is that U-235 occurs in nature, a larger amount of U-235 is required to make an explosion of equal yield to a plutonium weapon. Plutonium-239 (Pu-239) is more expensive to produce and must be made artificially, but it can be used to obtain a higher yield-to-weight ratio, smaller weapons size, and decreased weight.

From 1945-1947, the U-235/Pu-239 production ratio in the U.S. was approximately 8 to 1. It was therefore highly desirable to utilize U-235 and achieve the maximum efficiency in the use of both U-235 and Pu-239. Consequently, composite fission cores containing both U-235 and Pu-239 were developed; these fission cores were actually stockpiled at the end of 1947 for use in the Mark III implosion type bomb, although the percentage of plutonium needed to achieve maximum effect was then unknown. There is still some six to seven times more U-235 in the U.S. weapons stockpile than Pu-239 (see *Nuclear Weapons Databook, Volume II*). Most U.S. nuclear weapons contain both Pu-239 and U-235.

Critical Mass

Small amounts of fissile material will not sustain a chain reaction because a large fraction of neutrons leak out, making them unavailable to cause fission in other nuclei. The minimum mass of material necessary to sustain a chain reaction is called the *critical mass* and is dependent on the type of fissile material, its density, and its geometry. A mass that is less than the critical amount is said to be *subcritical*, while a mass greater than the critical amount to achieve a multiplying chain reaction is referred to as *supercritical*.

Because a sphere has the highest volume-to-surface ratio of any solid shape and, therefore, the least number of escaping neutrons per unit of material, it is the shape for which the critical mass is smallest. The critical mass of a bare sphere of U-235 at normal density is approximately 52 kilograms (kg), that of U-233 about 16 kg, and that of certain dense metallurgical phases of Pu-239 as low as 10 kg.¹⁸

The critical mass can be lowered in several ways. The fissile material may be surrounded by a shell of other material to reflect some of the neutrons which would otherwise escape. Practical *reflectors* can reduce the critical mass by a factor of two or three so that about 5-10 kg of either Pu-239 or U-233 and about 13-25 kg of U-235 at normal density can be made critical.¹⁹

The critical mass is also lowered if the material is compressed to increase its density.²⁰ Consequently, an efficient practical fission bomb, which depends on extremely high compression of the nuclear core, could use significantly smaller amounts of fissile materials

16 Weapon-grade plutonium is defined as plutonium containing less than 7 percent Pu-240; fuel-grade plutonium is 7 percent to less than 19 percent Pu-240; and reactor-grade is 19 percent or greater in Pu-240 content.

17 Pu-240 is the most troublesome of plutonium isotopes for bomb design, but not because of its inability to sustain a chain reaction. Pu-240, in fact, has a bare sphere fast neutron critical mass of 40 kg, less than that of U-235. Although when Pu-240 is mixed with Pu-239 it raises the critical mass of the mixture, the undesirable aspects of Pu-240 arise primarily from the fact that it fissions spontaneously with a much shorter fission half-life than Pu-239.

For one kilogram of U-235, spontaneous fission produces approximately one neutron per second. The spontaneous fission rates of weapons-grade plutonium are 60,000 and 300,000 times higher. Another [smaller] source of neutrons is the alpha-n reaction. In this case, radioactive decay of the fissile isotope yields alpha particles, some of which then collide with impurities such as boron, carbon, or oxygen to yield neutrons.

The classic problem presented by background neutrons is that of *preinitiation* of the nuclear-fission chain reaction. In order to assemble fissionable material to produce a nuclear explosion, a subcritical mass (or masses) of material must be rapidly moved into a configuration which has a level of supercriticality sufficient to produce a significant nuclear yield before it blows itself apart. Preinitiation in a nuclear explosive is defined as the initiation of the neutron chain reaction before the desired degree of supercriticality has been achieved. Because the nuclear yield depends upon the degree of supercriticality at the time the chain reaction is initiated, preinitiation will result in a lower yield. However, initiation is a statistical process and can be understood using statistical techniques.

Preinitiation, by itself, does not necessarily make an explosive unreliable. Preinitiation *does* result in a statistical uncertainty in the yield. Another way to state this is that the probable nuclear yield is statistically distributed between predictable upper and lower limits, which are likely to be more than a factor of 10 apart. For a well-understood design properly constructed, however, the most probable yield range could be predicted within much closer limits. [Mark, *op. cit.*, p. 141.]

Because of their particular sensitivity to preinitiation, gun devices (see Fission Weapon Design) are never designed with plutonium of any quality.

For low-technology [implosion] devices using high neutron background materials the probable yields could be lower by a factor of 3 to 10 or more (depending on the design) than using low-neutron background materials (i.e., U-233, U-235 and weapons-grade plutonium). Military useful weapons with reliable yields in the kiloton range can be constructed using low technology, (emphasis added) [Mark, *op. cit.*, p. 142.]

The U.S. tested a weapon constructed with reactor-grade plutonium in 1957. Had the first U.S. nuclear device (*Trinity* test, 16 July 1945) been constructed with reactor-grade plutonium its yield would have exceeded 1 kiloton. Using high-technology, or sophisticated design techniques, the problems presented by preignition can be largely overcome. Fuel-grade and reactor-grade plutonium are not used in U.S. weapons, but not primarily because of preinitiation. Pu-240 (and Pu-241) are more radioactive than Pu-239 and therefore generate more heat that must be dissipated if the integrity of the device is to be maintained for extended periods of time. Pu-240 is also more hazardous to handle than Pu-239, thus complicating further the manufacture of weapons using reactor-grade and fuel-grade rather than weapon-grade plutonium.

18 Pu in alpha phase ($M_c = 17$ kg for Pu in delta phase). The critical mass of Pu-239 is lower than that of U-235 because it has a higher fission cross-section—that is, each Pu-239 nucleus is more likely than a U-235 nucleus to capture a neutron and fission—and it produces on the average more neutrons per fission than U-235.

19 $M_c = 26.6$ kg for sphere of 94 percent U-235 surrounded by 1.74 inches natural uranium; $M_c = 8.4$ kg for sphere of alpha phase Pu-239 (4.5 percent Pu-240) surrounded by 1.6 inches natural uranium; and $M_c = 7.6$ kg for sphere of 98 percent U-233 surrounded by 2.1 inches natural uranium. Lower values of M_c can be achieved with other reflecting materials and/or thicker reflectors.

20 For a spherical mass of fissile material of radius, R , and uniform density, ρ :

$$(\rho \cdot R)_{\text{crit}} = \text{constant.}$$

If a fixed core mass, M , is uniformly compressed, the density is given by

$$\rho = 3M/4\pi R^3$$

consequently, the critical mass is approximately proportional to the reciprocal of the square of the density, i.e.,

$$M_{\text{critical}} = K/\rho^2.$$



Figure 2.2 Interior components of *Fat Man* type implosion bomb. The spherical shell of twelve pentagonal sections contains explosive "lenses" surrounding a uranium tamper and plutonium core.

than mentioned above. On the other hand, to obtain an appreciable fission yield more than one critical mass may be necessary. Thus, different types of nuclear weapons use different amounts of nuclear materials, and the reflected critical mass values discussed above—about 15 kg of U-235 and 5 kg of Pu-239—indicate only the order of magnitude of the actual amount of fissile material that may be required for a nuclear weapon. Most weapons in the U.S. arsenal are believed to use only a fraction of a critical mass (at normal density)—a "fractional crit"—as the fissile component.²¹ Other things being equal, fission weapons of higher yield require larger quantities of fissile material; therefore, the actual amount of fissile material in a weapon depends on the desired yield and the sophistication of the design.

Fission Weapon Design

An explosion is the release of a large quantity of energy in a small volume in a short period of time. There are numerous ways to assemble nuclear fissile materials to make them explode. For an efficient bomb or weapon the goal is to achieve a rapidly multiplying chain reac-

tion that within a very short time—a few microseconds—involves a very large number of atomic nuclei.

The fission chain reaction can be viewed as a sequence of stages or "generations," each marked by the fissioning of nuclei by neutrons produced in the preceding generation. The "generation time" is the average time between the emission of a fast neutron and its absorption by another fissionable nucleus, taking into account that neutrons are also lost by leakage and capture in other materials. The value of the generation time is roughly 0.01 microsecond (a "shake")²² and varies depending on the kind of fissionable material used, the design of the weapon, and the densities achieved during the explosion.²³

The energy release from a fission device takes place over a number of generations, depending on how many neutrons from fission in one generation remain to produce fissions in the next generation. In the fission of a single nucleus, between 2.5 and 3 fast neutrons are emitted. If, for example, 2 of these survive to produce other fissions, then the energy release from a device with a yield in the range 1 to 100 Kt would occur in about 53 to 58 generations.²⁴ Moreover, 99.9 percent of the energy release occurs in about the last 7 generations, which is roughly the last .07 microsecond of the explosion.

The energy release, or yield, is proportional to the number of nuclei fissioned, which is equivalent to the number involved in the chain reaction. To achieve a significant yield, the mass of the assembled fissile material must be several critical masses in order to obtain and maintain a rapidly multiplying chain reaction and avoid disassembly (i.e., becoming subcritical) before much material has been fissioned. This can be achieved by assembling two (or more) subcritical mass elements, or by changing the density and geometry of a fissile mass, initially subcritical, to reduce its critical mass. Higher yields (larger explosions) are obtained by design techniques which increase the mass above critical, and increase the time the fissile material is held together before the energy released by the nuclear explosion blows the weapon materials apart, stopping the chain reaction.

To keep the weapon from exploding and becoming subcritical before a reasonable fraction of its fissionable

21 The idea of using a fraction of a critical mass ("fractional crit") for an atomic explosion was originated by Hans A. Bethe from implosion calculations during the Manhattan project. After fission bombs had been thoroughly developed by postwar Los Alamos Laboratory the fractional crit became a practical possibility. It was strongly advocated by the Laboratory and the AEC in 1948-1949; see Hans A. Bethe, "Comments on the History of the H-Bomb," 1954, reprinted in *Los Alamos Science*, Fall 1982, p. 45.

22 Glasstone and Dolan, *op. cit.*, pp. 16, 17.

23 For a bare sphere critical assembly of U-235 the generation time is .66 shake. It is 2.0 shakes for a critical assembly with a U-235 spherical core and a thick natural uranium reflector, and .35 shakes for a bare sphere critical assembly of plutonium; John D. Orndoff, *Nuclear Science and Engineering*, 2, 450-460 (1957).

24 A yield of 1 Kt requires the fissioning of 1.45×10^{23} nuclei. If $1 + x$ neutrons per fission from one generation produce fissions in the next generation, the population of fission producing neutrons after n generations is given by $\exp(xn)$ for a chain initiated by a single neutron (Glasstone and Dolan, *op. cit.*). In the example in the text above, $x=1$. For *Little Boy* and *Fat Man*, the fission devices detonated over Japan, x is estimated to have been less than 0.7 neutrons per fission. For neutron leakage data, see George D. Kerr in V.P. Bond and J.W. Thiessen, eds., *Reevaluations of Dosimetric Factors, Hiroshima and Nagasaki*, *op. cit.*, pp. 64, 65.

nuclei undergoes a nuclear reaction, the fissile material is surrounded by a heavy material called a *tamper*. The same material may be used for both the tamper and the reflector; consequently, these terms are often used interchangeably. If constructed of fissionable material, e.g., U-238, the tamper will contribute to the fission yield as a result of the fissioning of its nuclei by the fast neutrons coming from the interior (see also Thermonuclear Weapon Design).

Two basic nuclear weapon design approaches that are used to achieve a supercritical mass of fissile material are the implosion technique and the gun assembly technique. In the *implosion technique*, a peripheral charge of chemical high explosive (HE) is uniformly detonated in a manner designed to compress (implode) a subcritical mass into a supercritical configuration. For example, an implosion device might consist of a spherical core of fissile material (a fraction of a critical mass) surrounded by a material that acts as both a tamper and a reflector; the tamper is then surrounded by high explosives. When detonated, the explosive sets up an implosion, or ingoing shock wave, that can create overpressures of millions of pounds per square inch in the core, increasing the density by a factor of two or more and thereby making the previously subcritical mass supercritical. The material occupying the volume within the surrounding high explosive charge, including the fissile core and the tamper/reflector, is commonly referred to as the "pit."

The implosion technique is commonly used in nuclear weapons where the fissionable material is Pu-239, U-235, or a composite of the two. It was used in the first U.S. nuclear test (*Trinity*, 16 July 1945) and also in *Fat Man*, the second nuclear weapon dropped on Nagasaki, Japan; the spherical assembly consisted of a thick charge of high explosive to compress a subcritical mass of Pu-239 surrounded by a thick layer of U-238 that served as both the tamper and reflector.

The *gun device* involves the assembly of two (or more) masses of fissionable material, each less than a critical mass. A conventional explosive is used to propel the subcritical pieces of fissionable material together, thereby assembling a supercritical mass. For example, a subcritical mass of fissionable material can be propelled down a tube, i.e., shot down a "gun barrel," into a second subcritical mass. This approach was used in *Little Boy*, the U-235 weapon dropped on Hiroshima, Japan. For that weapon, the second mass was held in a tamper consisting of a thin layer of tungsten carbide on the inside and steel on the outside. The W33 artillery fired atomic projectile now in the U.S. weapons stockpile

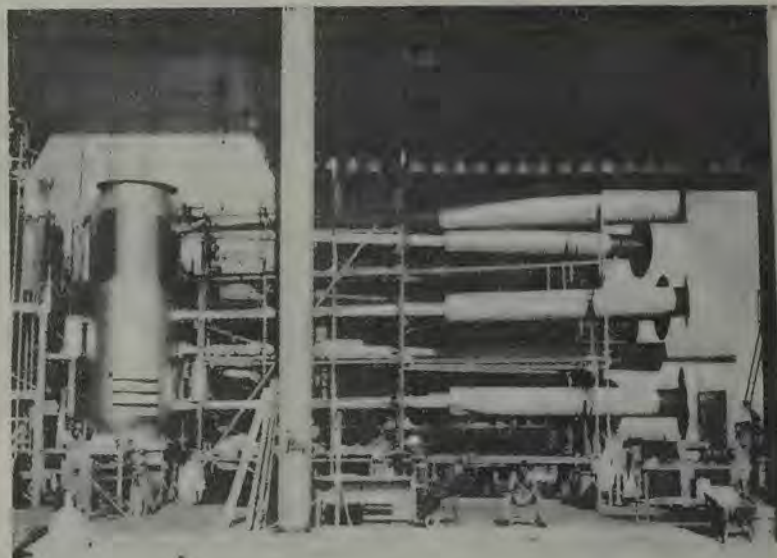


Figure 2.3 The *Mike* device, the first successful thermonuclear test, had a reported weight of 62 tons, due in part to the cryogenic equipment needed to maintain its thermonuclear fuel, deuterium, at liquid temperatures. Tested 31 October 1952 in Operation Ivy, at Elugelab, Enewetak Atoll, *Mike* had a yield of 10.4 megatons.

utilizes the gun assembly technique. Gun devices are conceptually relatively simple and can be designed with high confidence. Their design almost of necessity requires the use of U-235, rather than plutonium. Because an implosion device can generally be made with higher efficiency (higher yield for same amount of material) than a gun device, most fission devices in the U.S. stockpile, even those utilizing U-235 alone, use the implosion rather than the gun assembly technique.

Fusion Weapons

Fusion Reactions

"Thermonuclear" weapons, also referred to as "fusion" or "hydrogen" weapons, are usually defined as atomic weapons in which at least a portion of the release of energy occurs through nuclear fusion. The fusion reaction rates are extremely sensitive to temperature and are extremely small at normal temperatures. Only at 10-100 million degrees Kelvin—the interior of the sun is 14 million degrees Kelvin—are the rates sufficiently high to make fusion weapons (or reactors) possible; hence the term "thermonuclear." In thermonuclear weapons, the required temperatures and the required density of the fusion materials are achieved with a fission explosion.



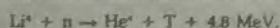
Figure 2.4 The Mk-17, the first U.S. droppable thermonuclear bomb to be tested, weighed 21 tons and had a yield in the megaton range.

Thermonuclear Fuels

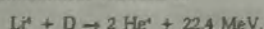
As noted previously, the D-T reaction is the principal source of fusion energy in thermonuclear weapons.²⁵ It is not necessary, however, to use elemental deuterium and tritium, which are gases at ordinary temperatures, directly in a thermonuclear weapon. The principal thermonuclear material is likely to be lithium-6 deuteride, which is a solid chemical compound at normal temperatures. In this case the tritium is produced in the weapon itself by neutron bombardment of the lithium-6 isotope during the course of the fusion reaction.²⁶ Since tritium decays radioactively (5.5 percent is lost each year),²⁷ lithium-6 deuteride has the added advantage of a longer storage life compared to tritium. Once fusion burn has been initiated, the action of fast neutrons on the isotope lithium-7, in the material lithium-7 deuteride, could be the source of additional tritium.²⁸

²⁵ A notable exception is the first full-scale American thermonuclear explosion (Mike Shot, Operation Ivy, Eniwetok Atoll, 31 October 1952), which used liquid deuterium and had a yield of 10.4 megatons (Mt), but this was a nuclear "device" designed for experimental purposes, not a prototype for an operational bomb.

²⁶ Tritium is bred from lithium-6 in the reaction:



When this reaction is combined with reaction (i) in footnote 4 of this chapter, the net thermonuclear reaction is:



²⁷ Tritium has a half-life of 12.33 years.

²⁸ $\text{Li}^7 + n \rightarrow \text{T} + \text{He}^4 + n.$

This tritium bonus was verified in 1954; see Lee Bowen, *A History of the Air Force Atomic Energy Program 1943-1953* (Washington: USAF Historical Division), Vol. IV, p. 40.

Thermonuclear Boosted Fission Weapons

By incorporating thermonuclear fuel, typically deuterium and tritium gas (or lithium hydrides) directly into (or proximate to) the core of fissile material, the efficiency of the fission bomb can be improved; that is, one can obtain a much higher yield from a given quantity of fissile material, or alternatively the same yield with a much smaller quantity. This process is called "boosting."²⁹ The fusion process itself may add only slightly to the yield of the device. Far more important to the yield is the extra quantity of free neutrons produced as a result of the fusion reaction.³⁰ These in turn produce additional fissions in the plutonium or uranium in the weapon, resulting in the increased efficiency. Thus, in boosted weapons, the thermonuclear fuel is used primarily as a source of neutrons to help the fission reactions, rather than as a direct source of yield. Boosted weapons are therefore basically fission weapons.

Because tritium decays radioactively, the effectiveness of the boosting process can degrade with time. Consequently, in stockpiled weapons which use tritium gas, the tritium is periodically replaced to ensure that a sufficient amount will be available.

Thermonuclear Weapon Design

In thermonuclear weapons, the fusion material can be incorporated directly into (or proximate to) the fissile core—for example in the boosted fission device—or external to the fissile core, or both. In the latter cases, radiation from a fission explosive is contained and used to transfer energy to compress and ignite a physically separate component containing the fusion material (and in some cases fissile material). The fissile core is referred to as the primary, and the component with the fusion material external to the primary is called the secondary. The weapon in this case would be said to have two stages.

The radiation from the fusion secondary can be contained and used to transfer energy to compress and ignite a third, or tertiary, stage, and the tertiary could similarly ignite a fourth, and so on. There is no theoretic-

²⁹ The boosting principle was recognized at least as early as November 1945 when possibilities of this general type were included in a patent application filed at Los Alamos. The designation "booster" only became general after its use by Edward Teller in September 1947; J. Carson Mark, "A Short Account of Los Alamos Theoretical Work on Thermonuclear Weapons, 1946-1950," LA-5647-MS, (Los Alamos: LASL, July 1974), p. 9. Teller in 1947 invented a "booster" design using liquid deuterium and tritium as the thermonuclear fuel. The design of this device was frozen in October 1950, and it was tested on 24 May 1951 in Shot Item of the Greenhouse series; J. Carson Mark, *The Bulletin of the Atomic Scientists*, March 1953, p. 47. The detonation of this 45.5 Kt device was a major contribution to the development of thermonuclear weapons. Shot George, an earlier detonation of this series (on 8 May 1951) produced the first significant U.S. thermonuclear reaction. George was an experiment using a fission bomb to ignite a small quantity of deuterium and tritium that contributed only a small amount to the 225 Kt yield.

³⁰ Complete fusion of 1 kg of D-T releases about 25 times as many free neutrons as the complete fission of 1 kg of uranium or plutonium. Alternatively, fusion produces up to 8 times more free neutrons than fission for the same energy.

Enhanced Radiation Weapons

cal limit to the number of stages that might be used and, consequently, no theoretical limit to the size and yield of a thermonuclear weapon. A thermonuclear weapon with a separate primary and secondary may, but does not necessarily, take advantage of boosting the primary.

While uranium-238 cannot maintain a self-sustaining fission explosion, it can be made to fission by an externally maintained supply of fast (highly energetic) neutrons from the fission or fusion reactions. Thus the yield of a nuclear weapon can be increased by surrounding the device with U-238, in the form of either natural or depleted uranium.³¹ This approach is particularly advantageous in a thermonuclear weapon where there is an abundance of fast neutrons from the fusion reaction. In a thermonuclear device, this U-238 blanket is sometimes referred to as the third stage of what would otherwise be a two stage weapon.

In general, the energy released in the explosion of a large thermonuclear weapon stems from three sources—a fission chain reaction, the first stage; “burning” of thermonuclear fuel, the second stage; and the fission of the U-238 blanket (if one exists), the third stage—with, very roughly, half the total energy stemming from fission and the other half from fusion. However, to obtain tailored weapons effects or to meet certain weight or space constraints, different ratios of fission-yield-to-fusion-yield may be employed, ranging from nearly pure fission yield weapons to a weapon where a very high proportion of the yield is from fusion.

Enhanced Radiation Weapons (Neutron Bombs)³²

The “neutron bomb” is a thermonuclear device designed to maximize the lethal effects of high energy neutrons produced by the fusion of deuterium and tritium and to reduce the blast (the kinetic energy of charged particles) from the explosion. In this weapon

the burst of prompt nuclear radiation (neutrons and gamma rays) is enhanced by minimizing the fission yield relative to the fusion yield. This is accomplished, in part, by the elimination of (or substitution for) U-238 components, particularly the U-238 tamper and blanket. Thus, the neutron bomb is referred to as an “enhanced radiation” (ER) weapon.

The neutron bomb uses deuterium with tritium, rather than lithium-6 deuteride, as the fusion material to maximize the release of fast neutrons. Each reaction of one deuterium nucleus with a tritium nucleus gives rise to a neutron with an energy of approximately 14 million electron-volts (or a “14 MeV neutron”). Such neutrons from fusion are up to six times more numerous per kiloton of energy release than are neutrons escaping from the fission chain reaction in a fission bomb.³³

Consequently, for a given explosive energy release, it is possible for a lethal dose of nuclear radiation to be delivered on the battlefield for a somewhat greater distance in the case of a neutron bomb than for a fission weapon. The lethal radius of a neutron bomb's radiation dose for a low altitude burst is about 700 meters. This is about twice the lethal radius of a fission weapon with an equal yield and about the same lethal radius as a fission weapon with ten times the yield.³⁴

For intermediate and large yields, the destructive radius of the blast generally far exceeds that of nuclear radiation. The blast damage radius drops off more rapidly with decreasing yield than the lethal radius of the radiation. Only in the energy range of 1 Kt and below would the radiation kill radius of an ER weapon (due to high-energy neutrons) considerably exceed the destructive radius of the same weapon due to blast.³⁵ Although a 1 Kt ER weapon has a lethal radius for nuclear radiation equal to that of a 10 Kt fission weapon, the fission weapon has a considerably greater radius for blast damage to urban structures.³⁶

31 While other fissionable materials such as U-234, U-236, and Th-232 could also be used, U-238 is used because it is readily available as tails from the enrichment plants (in quantities far more plentiful than U-234 or U-236), and it has a higher fission cross section than Th-232.

32 See Richard L. Garwin, “Trends in the Technological Development of Nuclear Weapons Systems,” Third Draft of Chapter 2 for 1980 UN Report on Nuclear Weapons, 27 March 1980, pp. 10-11; Herbert Scoville, Jr., “The Neutron Bomb,” *The Arms Race and Arms Control*, SIPRI, 1982, p. 115; and S.T. Cohen, *The Neutron Bomb* (Cambridge, Mass.: Institute for Foreign Policy Analysis, Inc., November 1978), p. 66 ff.

33 The energy released from a typical (low-yield) fission bomb consists of 50 percent blast, 35 percent thermal radiation (heat), 5 percent prompt radiation (gamma-rays, x-rays, and neutrons), and 10 percent residual radiation (fallout); whereas an ER weapon consisting of half fission and half fusion releases energy as 40 percent blast, 25 percent thermal, 30 percent prompt, and 5 percent residual radiation; Fred Kaplan, “The Neutron Bomb,” *The Bulletin of the Atomic Scientists*, October 1981, p. 6.

34 Estimates used by the U.S. Army (and the Joint Chiefs of Staff) are that a dose of 8000 rads is necessary to inflict “immediate permanent incapacitation” on a human being. The “military radius” (or radiation-kill radius) is the distance from ground zero inside of which the dose is greater or equal to this value. Following are the dose radii for tank crews for doses of 8000 and 650 rads. At 650 rads personnel will become functionally impaired within 2 hours of exposure. Radii are in meters for a burst height of 150 meters. [Cohen, *op. cit.*; Scoville, *op. cit.*]

Weapon	Radius (m)	
	8000 rad	650 rad
1 Kt fission	360	690
1 Kt ER	690	1100
10 Kt fission	690	1100

35 Theoretically, the pressure at a given distance from an explosion is proportional to the cube root of the energy yield; see Glasstone and Dolan, *op. cit.*, p. 100. Thus, a doubling of the blast kill radius results from an 8-fold increase in yield; the same 8-fold increase of yield adds only about 400 m to the prompt lethal radius of nuclear radiation effects for a neutron bomb. The difference in scaling occurs because the neutrons and gamma rays are strongly absorbed in the atmosphere while the blast wave is only slightly attenuated.

36 A pressure of 4 psi (pounds per square inch) produces a moderate level of blast damage to urban structures. Lethal radiation and blast damage radii in meters for low altitude burst (150 meters) are as follows [Cohen, *op. cit.*, and S.T. Cohen, *Strategic Review*, Winter 1978, p. 9]:

Weapon	Radius (m)	
	8000 rad	4 psi
1 Kt ER	690	550
1 Kt fission	360	810
10 Kt fission	690	1220

A 1 Kt neutron weapon is more costly to manufacture than a 10 Kt fission weapon. A neutron weapon also has more constraints on its delivery, in part because of the need to maintain the tritium supply.

Third Generation Weapons

A new "third generation"³⁷ of advanced nuclear weapon concepts is now emerging.³⁸ These weapons are described as being highly selective in their effects and suited to "purely defensive"³⁹ use for destroying an adversary's offensive systems. Development is being led⁴⁰ by the Lawrence Livermore National Laboratory⁴¹ with participation of other weapons laboratories. The new designs are to be ready for introduction in the 1990s.⁴²

Among third-generation concepts are the following:

- An x-ray laser pumped by x-rays from a nuclear explosion for use as a defense against incoming enemy ballistic missiles above the earth's atmosphere, or as an anti-satellite weapon.⁴³
- Enhanced radiation (ER) weapons of very low yield (50-100 tons) guided by radars into the path of incoming ballistic missiles high in the atmosphere. Compared to these weapons, the ER weapon for battlefield use is characterized by officials as a "crude forerunner"⁴⁴ of a third-generation ER weapon; the ER warhead of the SENTRY anti-ballistic missile system (now under intensive development at Livermore) is described as a device "at the beginning of the defensive use of nuclear weapons."⁴⁵
- EMP weapons specially designed to create a large electromagnetic pulse (EMP) to burn out enemy communications, utilizing directed or non-directed EMP created by a nuclear explosion above the atmosphere.



Figure 2.5 B-52 sits atop the TRESTLE electromagnetic pulse simulator at Kirtland Air Force Base, New Mexico. Ten million volts of EMP energy are created to simulate effects from nuclear explosions on aircraft and electrical equipment.

The realm of advanced technologies now includes a directed form of EMP using a "high power microwave coherent beam of immense peak power." The Department of Defense Directed Energy Program (under DARPA) includes "radio frequency weapons" as one of its three major programs.⁴⁶

The feasibility of reliable "defensive" nuclear weapons has been questioned by some familiar with weapons design. Skepticism about the use of the x-ray laser as a new weapon has been particularly keen.⁴⁷ One critic has noted that "in the nuclear area, the offense will continue to have the advantage and can negate any defensive weapon with relatively little effort."⁴⁸ It has been suggested that should a defensive breakthrough occur, "it will probably not involve either nuclear fission or fusion in an essential way."⁴⁹

For both ER and fission weapons, prompt radiation is the predominant battlefield effect. The radiation-kill radius for tank crews is significantly larger for both the 1 Kt ER and 10 Kt fission weapons than the radius at which the tanks themselves will be damaged by blast. (For a discussion of battlefield strategies see Scoville, *op. cit.*)

³⁷ The first generation was in the 1940s and 1950s when the first nuclear bombs and warheads were developed. The second generation came in the 1960s with the development of compact high-yield warheads.

³⁸ Judith Miller, *New York Times*, 29 October 1982; *FAS Newsletter*, October 1982, p. 6.

³⁹ Edward Teller, *Science*, 24 December 1982, p. 1270.

⁴⁰ *Ibid.*

⁴¹ As stated in the Directors' Comments of the FY 1982-1987 Lawrence Livermore Institutional Plan (1982, p. 2, later recalled), "Increased demand for modern nuclear weapon systems, coupled with significant developments in the design of a new nuclear weapon for ballistic missile defense applications and a concept for directing the energy of a nuclear weapon, strengthens our recommendation for additional attention in the areas of nuclear design and testing."

⁴² Dr. Richard Wagner, HASC 97-41, April 1982.

⁴³ The nuclear pumped x-ray laser was reported to have been successfully tested underground at the Nevada Test Site by Lawrence Livermore scientists; see *AW&ST*, 23 February 1981.

⁴⁴ Quote from Judith Miller, *New York Times*, *op. cit.*

⁴⁵ Edward Teller as quoted in *FAS Newsletter*, *op. cit.*, p. 9.

⁴⁶ Letter of Charles E. Kinney, *Science* 83, March 1983, p. 21.

⁴⁷ For example, by Richard Garwin, as reported by David Perlman, *San Francisco Chronicle* 25 September 1982. Hugh E. DeWitt states that a variety of countermeasures could make the x-ray laser ineffective. For example, the enemy ballistic missile could release large pieces of metal chaff having the same radar image as the missile itself; see Hugh E. DeWitt, Lawrence Livermore National Laboratory, in letter to David Saxon, President, University of California, 3 October 1982.

⁴⁸ Hans A. Bethe, *Science*, 24 December 1982, p. 1270. Bethe is reported to have later said, after reviewing theoretical designs and studies for the x-ray laser at LLNL, that "this is the one and only one proposal that makes scientific sense," but "to translate this into an operational device is a fantastic business"; R. Jeffrey Smith, *Science*, 1 July 1983, p. 30.

⁴⁹ Herbert F. York, *Science*, 21 January 1983, p. 236.

Warhead Features

Warhead Features

A nuclear weapon consists of a nuclear warhead package of fissile and/or fusion materials. Typically, it also contains a sequencing microprocessor; chemical high explosives; a neutron actuator; an arming system, the component which serves to ready (prearm), safe, or resafe (disarm) the warhead; a firing system; a fuzing mechanism (radar, pressure sensitive, time) which regulates the detonation of the warhead; and control and safety devices; all of which may or may not be an integral part of a warhead package, depending on its design and age.

Warhead Safety and Control

Depending upon its age and deployment location, each warhead contains safety and control devices which are to prevent accidental, unauthorized use (prearming, arming, launching, firing) or inadvertent use of nuclear weapons. In some cases, the safeguard and arming features are no more than a wire seal, a switch, and a lock; however, the modern weapons (and all weapons deployed outside the United States) have a series of complementary features, including locked containers and code controlled arming and fuzing systems. In addition, the "two-man rule"—which requires a minimum of two authorized personnel (each capable of detecting incorrect or unauthorized procedures) present during any functions where people come in contact with nuclear weapons or code materials related to their release—enforces adherence to safety and security procedures.⁵⁰

The most significant control feature of the warhead, the Permissive Action Link (PAL), is the incorporation of a coded switch or lock device in the arming line.⁵¹ A PAL is a coded "lock" that requires the insertion of a proper number (manual combination or electronic digits) in order to "open" (unlock) circuits to arm the weapon.⁵² With the newer PALs, after the repeated unauthorized entry of false numbers, the weapon locks, certain key components are made unusable, and the warhead would have to be returned to the assembly plant for repair.⁵³ Each typical PAL device (Category D) costs about \$50,000.

Each successive generation of PAL devices since their introduction—the latest is the "Category F" PAL—has represented more than just an increase in safety. The evolution of PAL from the original single code combination lock device for warheads (still deployed on some

Table 2.1
Warhead Safeguard Features

Command Disable ¹	inertial nonviolent code activated disabling device which destroys critical warhead components, rendering the warhead useless, integrated into the storage container
Electrical Safety	surge capacitors, exclusion region, for weak link/strong link system
Environmental Sensing Device (ESD)	barometric pressure sensor for arming bombs and artillery projectiles
Permissive Action Link ²	mechanical combination lock
Category A	
Category B	ground and airplane operated 4 digit coded switch, later version with limited try
Category C	single 6 digit coded switch, with limited try
Category D	multiple code 6 digit coded switch, with limited try
Category F	multiple code 12 digit coded switch, with limited try
Unique Signal Generator	arming-safing signal encoder for firing system requiring unique electrical signal
Weak Link/Strong Link	exclusion region for warhead electrical system preventing inadvertent or accidental electrical surges and firing

1 ACDA, FY 1978 ACIS, p. 96; ACDA, FY 1979 ACIS, p. 121; SASC, FY 1978 ERDA, p. 109.

2 ACDA, FY 1979 ACIS, pp. 92, 130; SASC, FY 1981 DOE, pp. 74-76.

older weapons) to the present 12 digit electronic "multiple-code coded switch" made it possible to control the release of individual warheads and specified yields, each with a uniquely generated code. Thus the "safety" features also serve to support the evolving limited nuclear war strategies.

The other safety and control systems incorporated into the warheads or containers besides the PAL—command disable feature, electrical safety, weak link/strong link, unique signal generator, etc.—are described in Table 2.1, Warhead Safeguard Features. Many newer weapons also contain "insensitive high explosives" (IHEs), a more stable chemical compound than previously used high explosives to "guard against detonation from fire, small arms, aircraft crash, or inadvertent release in flight."⁵⁴ IHEs, which are more expensive and

50 U.S. Army, *Operations for Nuclear Capable Units*, FM 100-50 (March 1980), p. 11.

51 SASC, FY 1982 DOE, pp. 266-267; background is also provided in *Military Applications of Nuclear Technology*, Part 1, pp. 44ff.

52 ACDA, FY 1979 ACIS, p. 92; ACDA, FY 1978 ACIS, p. 96.

53 ACDA, FY 1979 ACIS, p. 136.

54 ACDA, FY 1979 ACIS, p. 92.

heavier than previously used high explosives, will only be integrated into weapons which are transported frequently. According to the Sandia Corporation, "insensitive high explosives minimize the risk of detonation ... thereby reducing the risk of scattering of plutonium."⁵⁵ The IHEs, first incorporated into the B61 Mod 3 and 4 bombs, are now in the W80 cruise missile warhead and programmed for most future warheads.⁵⁶

Selectable Yields

There is little in the open literature describing how the yields of current nuclear weapons are varied. The actual designs probably incorporate one (or more) of the following procedures.

In a pure fission device, the yield can be varied by varying the timing of the initiation of the chain reaction

or by interchanging pits. In modern weapons, the chain reaction is initiated by a neutron gun.

In a boosted fission weapon, or a thermonuclear weapon with a boosted primary, the yield can be varied readily by carefully selecting the amount of tritium gas bled into the fissile core from an external reservoir.⁵⁷

In thermonuclear weapons with one (or more) fusion stages, the yield is varied by tritium control or by interchanging the pits. In addition, mechanical measures that dictate whether or not additional fusion stages ignite could be used, although there is no evidence in the open literature that this is done. The yield of some older bombs in the U.S. stockpile, e.g., the B28, appears to be varied by interchanging the pits. This procedure, which is performed on the ground, is less flexible than selecting the yield by tritium control, the approach used in more modern weapons.

Weights and Yields of Nuclear Weapons

The first nuclear explosive device, called the *Gadget*, had a plutonium core weighing about 13.5 lb (6.1 kg) which was imploded by some 5000 lb of high explosive.⁵⁸ The *Gadget* was tested at the Trinity Site's 100 foot tower in Alamogordo, New Mexico (16 July 1945,



Figure 2.6 The 100 foot tower at ground zero of the Trinity Site at Alamogordo, New Mexico. It was built to test the "Gadget" contained [without its plutonium core] in the crate at the base of the tower.



Figure 2.7 The "Gadget"—prototype for the *Fat Man* weapon design—at the top of the tower at the Trinity Site shortly before its test on 16 July 1945.

⁵⁵ SASC, FY 1981 DOE, p. 74.

⁵⁶ HASC, FY 1980 DOE, p. 140.

⁵⁷ Both tritium gas and deuterium gas in a selected volume ratio could be bled into the weapon's "pit" during the arming sequence.

⁵⁸ Major Gen. Leslie R. Groves, Memorandum for the Secretary of War, 18 July 1945, TOP SECRET (DECLASSIFIED) reprinted as Appendix P to Martin J. Sherwin, *A World Destroyed*, (New York: Alfred A. Knopf, 1975); see also James W. Kunetka, *City of Fire*, (Englewood Cliffs, N.J.: Prentice-Hall, Inc., 1978), p. 164.

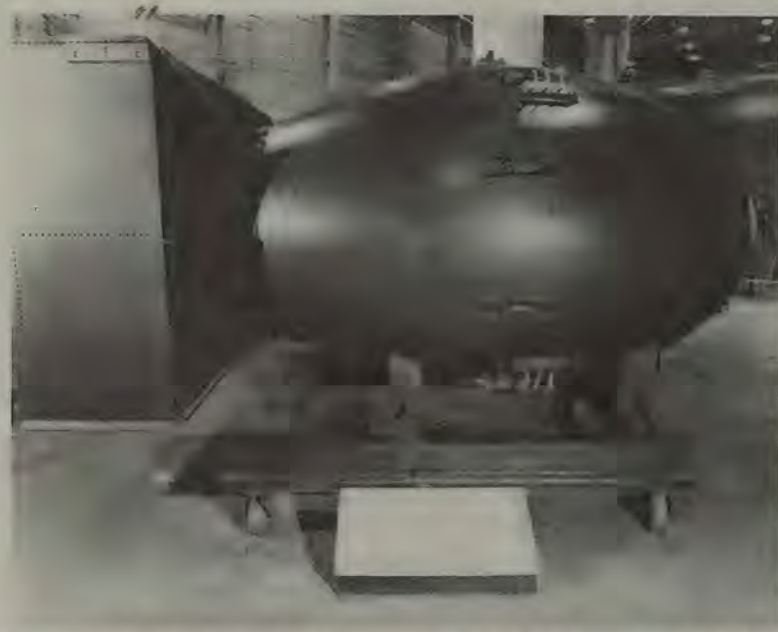


Figure 2.8 Plutonium fission weapon of the *Fat Man* type. *Fat Man* was the atomic bomb that was dropped on Nagasaki, Japan on 9 August 1945. Its yield was 22 kilotons.

5:29 am), with a yield of 22 ± 2 Kt.⁵⁹ The following description is from Kunetka.⁶⁰ "While the actual size of the plutonium core was about that of a grapefruit, the uranium tamper and explosive charges added considerably to the bomb's size. The core, tamper, and high explosives were held in place by a metal sphere made of twelve pentagonal sections. These were bolted together to form a sphere . . ."

Fat Man, the nuclear weapon which was dropped on Nagasaki, Japan (estimated 503 meters or 1650 ft,⁶¹ 9 August 1945, 11:02 am), was based on the same design as the *Gadget*; the weapons' names are often interchanged. *Fat Man* included stabilizing fins and a protective egg-shaped outer shell, or bomb casing, 60 inches in diameter. The bomb was 12 feet long and weighed 10,800 lb with fins.⁶² The yield of the *Fat Man* has been well-established at 22 ± 2 Kt.⁶³

Little Boy, the nuclear weapon dropped on Hiroshima, Japan (estimated 580 meters or 1903 ft,⁶⁴ 6 August 1945, 8:15 am), contained 60 kg of highly enriched uranium and utilized the gun assembly technique.⁶⁵ The



Figure 2.9 Nuclear weapon of the *Little Boy* type. *Little Boy* was dropped over Hiroshima, Japan on 6 August 1945, the first nuclear weapon ever delivered. It was 120 inches long, 28 inches in diameter and weighed 9000 pounds.

gun barrel had a diameter of over 6 inches, was 6 feet in length, and weighed about one-half ton (or less than one-fifth the weight of standard guns of that calibre). The bomb with its casing was 10 feet long, 28 inches in diameter, and weighed 8900 lb.⁶⁶ The yield of the Hiroshima *Little Boy* is uncertain, with estimates ranging from 12 Kt to 15 Kt.⁶⁷

The explosive yields of the first two implosion devices, the *Gadget* and *Fat Man*, both about 22 ± 2 Kt, corresponded to an efficiency⁶⁸ of about 17 percent in utilization of the 6 kg of plutonium,⁶⁹ whereas the 12 to 15 Kt *Little Boy* gun device had an efficiency of only about 1.3 percent. In order to achieve these efficiencies in the first U.S. designs, considerable weight was allocated to the chemical explosive and tamper. The yield-to-weight ratios were 0.0045 Kt/kg for the *Fat Man*, and 0.003 Kt/kg for the *Little Boy*, both very low compared to modern designs (see Figure 2.15).

The Mark III (based on *Fat Man*) was the only stockpiled implosion bomb from 1945 to 1948. In late 1948 the Mark IV assembly began entering the stockpile. Unlike

59 Reevaluation by D. Eilers; see John Malik in V.P. Bond and J.W. Thiessen, eds., *Reevaluation of Dosimetric Factors, Hiroshima and Nagasaki*, CONF-810928 (DE 81026 279) (Washington: U.S. DOE, Technical Information Center, 1982), p. 100. Nevada Operations Office gives 19 Kt from an earlier evaluation. "Announced United States Nuclear Tests July 1945-December 1981," NVO-209 Rev-2 (U.S. DOE, Office of Public Affairs, Nevada Operations Office, January 1982).

60 Kunetka, *op. cit.*, p. 123.

61 Malik, *op. cit.*, p. 105.

62 Robert T. Duff, Director, Office of Classification, DOE, letter to David A. Rosenberg, 4 December 1980.

63 Malik, *op. cit.*, pp. 98, 100, 107. Nevada Operations office gives 23 Kt based upon an earlier evaluation; see U.S. DOE, Nevada Operations Office, *op. cit.*, p. 22.

64 Malik, *op. cit.*

65 John McPhee, *The Curve of Binding Energy* (New York: Farrar, Strauss and Giroux, 1974), p. 14.

66 Duff, *op. cit.*

67 Malik, *op. cit.*, pp. 98, 107. Malik gives values of 12 ± 4 Kt deduced from blast effects, 15 ± 3 Kt from thermal effects and a "suggested best value" of 15 ± 3 Kt; Kerr (*op. cit.*, p. 88) prefers an earlier value of 12.5 ± 1 Kt; U.S. DOE, Nevada Operations Office, *op. cit.*, quotes a value of 13 Kt. "[Robert] Oppenheimer gave the *Little Boy* a good chance of 'optimal performance'; only a 12 percent chance of less than this; a 6 percent chance of an explosion under 5000 tons; and a 2 percent chance of one under 1000 tons of TNT"; Memorandum from Robert Oppenheimer to General Leslie Groves and William Parson, 23 July 1945; quoted by Kunetka, *op. cit.*, p. 178.

68 Since complete fission of 1 kg of Pu or U corresponds to 17.5 Kt,

$\epsilon = Y/17.5M$

where ϵ = efficiency, Y = yield in Kt and M = mass in kg.

69 For *Fat Man* it is assumed that 80 percent of the yield was from fission occurring in the Pu-239 core and 20 percent in the U-238 tamper. See Kerr, *op. cit.*, p. 81.



Figure 2.10 The *Davy Crockett*, the smallest yield nuclear weapon ever deployed by the U.S., on display at the National Atomic Museum. It is 31 inches in length (25.5 inches excluding fins) and 11 inches in diameter.



Figure 2.12 The Mk-23, a 16-inch projectile for Navy battleship guns, was the first and only Navy nuclear artillery shell.



Figure 2.11 The Mk-5 nuclear weapon on display at the National Atomic Museum. The Mk-5 was the first light-weight (3000 pounds) tactical gravity bomb.

the Mark III, the fissile core of the Mark IV could be inserted in flight. Also, from 1945 to 1951 the only gun-type weapons assembly in the stockpile was the *Little Boy*.⁷⁰

Most of the early explosions of other nuclear weapons nations have been in the same 10 to 20 Kt range as *Fat Man* and *Little Boy*. The first designs were undoubtedly conservative, with a sufficient amount of fissile material to assure a good yield without placing undue demands on the implosion or gun device used.⁷¹

The smallest nuclear weapon (in terms of size, weight, and yield) in the U.S. stockpile is the W54, a fission implosion weapon which is currently deployed as the warhead in the Special Atomic Demolition Munition (SADM). It was also utilized as the warhead for the FALCON missile (retired 1972), and the DAVY CROCKETT rocket (retired 1971). The reported yield of the DAVY CROCKETT was 0.25 Kt. The SADM may have yield options even smaller, i.e., in the 0.01 Kt range. The DAVY CROCKETT weighed 51 lb (23.2 kg), only 0.5 percent of the weight of the *Fat Man* bomb, and its dimensions were 30.97 inches (25.5 inches excluding fins) in length and 11 inches in diameter; in comparison, the *Fat Man* was 120 inches in length and 62 inches in diameter.⁷²

⁷⁰ David Allen Rosenberg, "U.S. Nuclear Stockpile 1945 to 1950," *The Bulletin of the Atomic Scientists*, May 1982, p. 26.

⁷¹ If substantially less than a critical mass of material at normal pressure is used, a much greater burden is placed on the performance of the chemical implosion device to achieve the compression needed for reasonable efficiency. In other words, the risk of a fizzle, or at least of a very inefficient explosion, is increased if there is an attempt to conserve fissile material in a first test.

⁷² U.S. Army, "Operation and Employment of the DAVY CROCKETT Battlefield Missile," XM-28/29, FM 9-11, June 1963. The weight of the SADM may be somewhat heavier. The DNA, "Motion Picture Catalog" January 1961, for example, lists the weight of the W54 as 26.6 kg; the W54 warhead was based on a design by Theodore Taylor at LASL and has also been referred to as weighing "less than 50 lbs," in McPhee, *op. cit.*, p. 8.



Figure 2.13 The Army's **280mm atomic cannon**, which was designed to fire the Mark-19, the first nuclear artillery shell.

The largest fission device ever detonated, the Super Oralloy Bomb, had a yield of 500 Kt.⁷³ This is larger than most *thermonuclear* weapons currently in the U.S. stockpile.⁷⁴ At yields over approximately 50 Kt, thermonuclear weapons can be produced at much lower cost and much less weight than pure fission weapons. Consequently, one does not expect to see high yield fission weapons stockpiled.

The first successful test of a thermonuclear device was on 31 October 1952 with the 10.4 Mt Mike shot at Enewetak Atoll.⁷⁵ The largest thermonuclear device tested was reportedly 58 Mt, a Soviet atmospheric test at approximately 12,000 feet at Novaya Zemlya, on 30 October 1961.⁷⁶

As noted previously, only the practical considerations of weight and size limit the yield of deliverable thermonuclear weapons. The first deliverable thermonuclear warhead in the U.S. stockpile was the B17 bomb (retired in 1957) which weighed almost 21 tons and had a yield of several megatons.⁷⁷ The largest warhead presently in the U.S. stockpile is the B53 bomb (and W53 warhead on the TITAN II missile) which has a yield of 9 Mt and a weight of 8850 lb, or a yield-to-weight ratio of about 2.2 Kt/kg, some 400 times that of *Fat Man*.

⁷³ Defense Nuclear Agency, "The Radiological Cleanup of Enewetak Atoll" (Washington, D.C.: 1981), p. 49. Event King, during Operation Ivy, 15 November 1952 (air drop detonation at 1500 feet). The Super Oralloy Bomb was designed by Theodore Taylor at LANL; McPhee, *op. cit.*, p. 8.

⁷⁴ For example, the thermonuclear W78 warhead—used on 300 MINUTEMAN III ICBMs—has a yield of 335 Kt, the highest yield of any currently deployed U.S. strategic missile, except for the 9 Mt TITAN II (W53) warheads which are soon to be retired.

⁷⁵ U.S. DOE, Nevada Operations Office, *op. cit.* The weight of the Mike device has been reported as 62 tons; Neyman and Sadilenko, *op. cit.*, p. 20.

⁷⁶ Herbert F. York, *The Advisors* (San Francisco: W.H. Freeman and Company, 1976), p. 93; see also Glasstone, 1964 Ed., Appendix B, p. 681A.

⁷⁷ National Atomic Museum, Albuquerque, NM. See Table 1.4 for list of U.S. warheads; the B17 may have been tested as the *Cherokee Shot* in Operation Redwing (airdrop 4320 feet, at Bikini Atoll, 20 May 1956). The *Cherokee Shot* was the first air drop by the U.S. of a thermonuclear weapon. See U.S. DOE, *op. cit.*

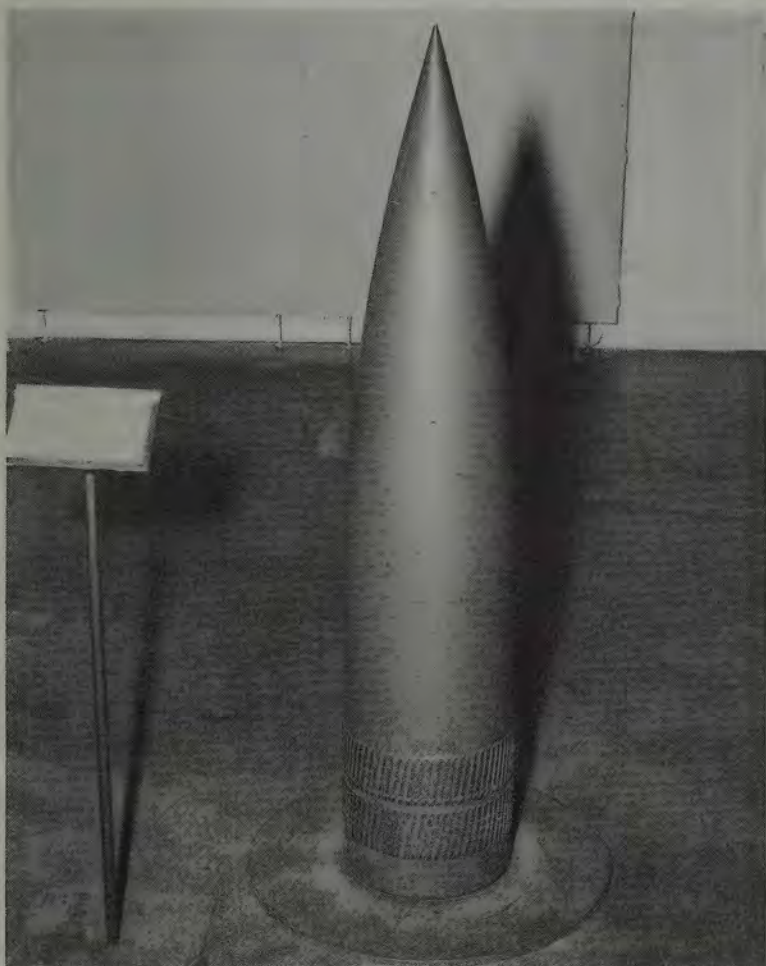
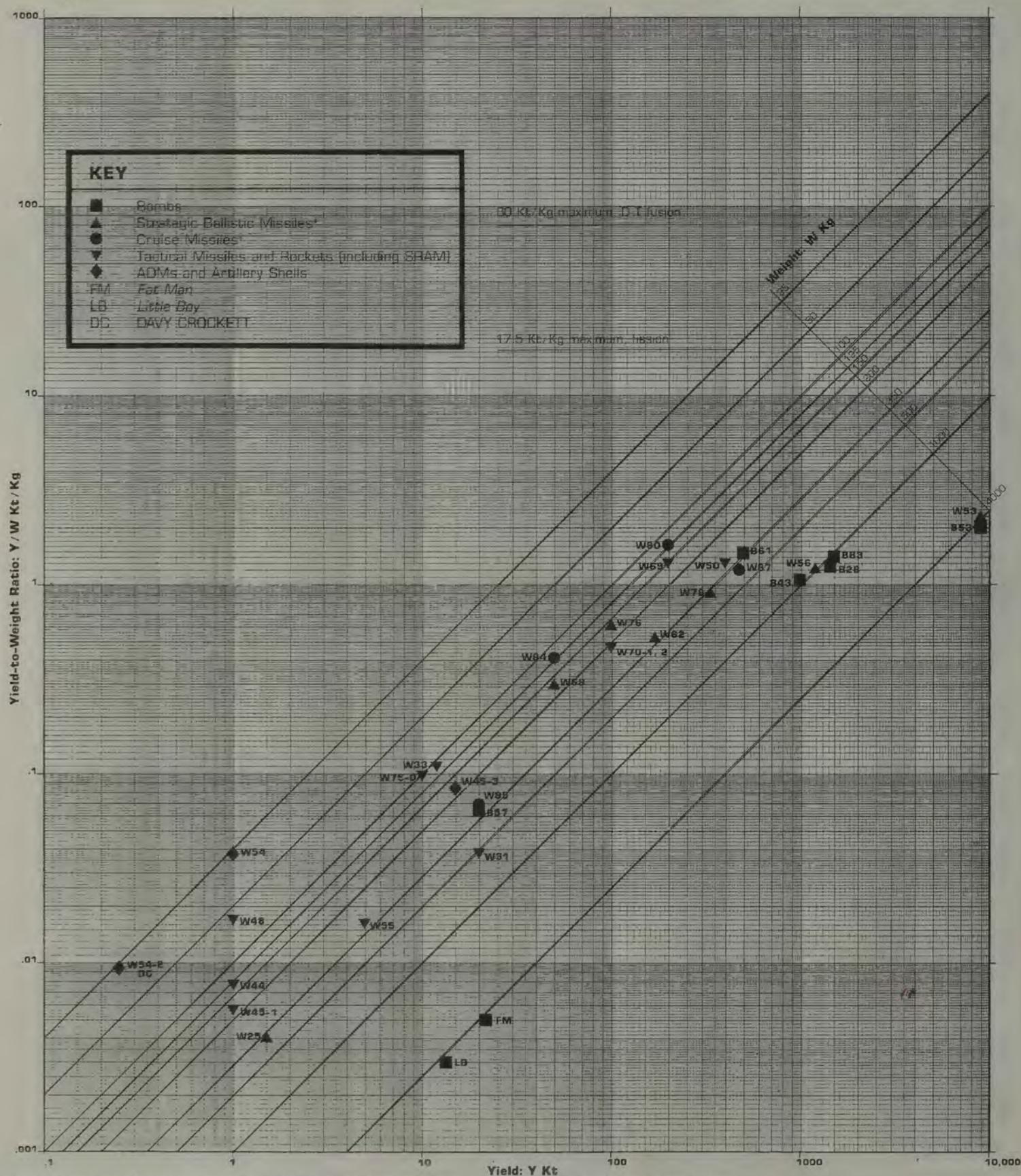


Figure 2.14 W19, a projectile for the Army's 280mm howitzer, was the first nuclear artillery shell. It was an oralloy weapon utilizing the gun assembly technique.

Modern thermonuclear weapons with yields above 100 Kt can be expected to have yield-to-weight ratios in the 1-3 Kt/kg range. This is still far below the theoretical limit of 80 Kt/kg represented by the complete fusion of deuterium-tritium material.⁷⁸ Figure 2.15 shows a plot of yield-to-weight ratio versus both yield and weight for nuclear warheads and bombs currently in the stockpile. The strategic missiles and bombs with yields greater than about 100 Kt (concentrated in the upper right portion of the figure) show yield-to-weight ratios in the range 0.3-2.5 Kt/kg. The low-yield tactical nuclear weapons (in the lower left portion of the figure) all have substantially smaller yield-to-weight ratios in the 0.004 to 0.1 Kt/kg range.

⁷⁸ See discussion of fusion energy release earlier in this chapter.

Yield-to-Weight Ratios



1 Warheads shown were in the nuclear stockpile as of 1982, with the following exceptions: Retired: W54-2 DAVY CROCKETT [DC], Little Boy [LB], Fat Man [FM]. Planned additions: W84 GLCM, W85 PERSHING II, W87 MX, and B84 Modern Strategic Bomb.

2 Maximum yields are used where warheads have variable yields. Data for yields and weights are subject to some degree of uncertainty.

3 Bomb weight includes the bomb casing.

4 Ballistic missile weight consists of the warhead package with the reentry vehicle and a fraction of the bus.

5 For cruise missiles, weight is for warhead package only.

Figure 2.15 Yield-to-Weight Ratios vs. Yield and Weight for Current Stockpile Nuclear Weapons and Bombs.^{1,2}